MotionPlay

Camera-Based Body-Motion Controller

AI Capstone: Milestone 1 Check-In (Week 3)

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Project Charter

The Problem & Goal

Conventional gameplay interfaces exclude users seeking accessible, highly active, or physical controller configurations.

Meanwhile, powerful depth camera sensors in modern smartphones remain deeply underutilized for local workspace spatial control.

Strategic Goal: Create an offline system utilizing a phone camera to track 3D skeletons, evaluate 5 key gestures, and emulate live gamepad outputs.

Scope Confirmation

Establishing rigid parameters for model and software iteration:

- **In-Scope:** iOS ARKit joint coordinate streams, user-recorded database (~1,000 motion clips), ML/DL classifiers, mapping servers.
- **Out-of-Scope:** Facial expression capture, finger configuration tracing, multiplayer gaming, cloud streaming, custom engines.

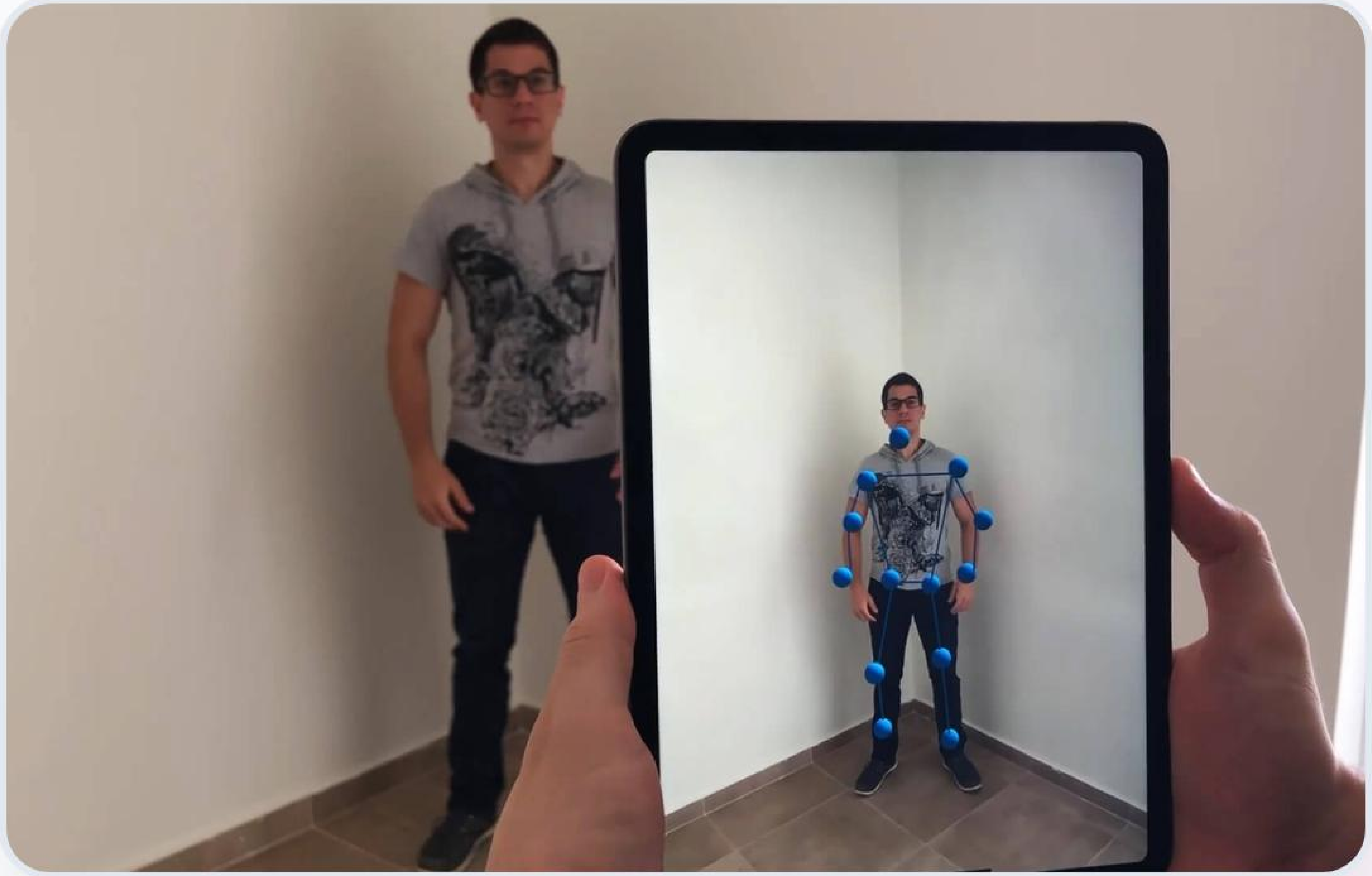
Data Acquisition

Our custom-built iOS tracking app leverages Apple's native **ARKit body tracking model** to capture 3D joint configurations directly, eliminating storage and frame latency.

Real-time joint trajectories are modeled spatially as absolute coordinates:

$P_{joint} = xyz$ with rotation angles $q = xyzw$

Pipeline Benchmark: ~1,000 dynamic sweeps tracked across 4 team developers (5 gestures x 50 repetition intervals).



Preprocessing Pipeline

Joint Filtering

Prune peripheral hand nodes and facial tracking. Isolate 20 core torso-limb joint points for robust translation.

Normalization

Root all absolute coordinates at the Hips. Scale dynamic bodies based on skeleton bone lengths for general bounds.

Resampling

Mitigate wireless connection stutter. Standardize coordinate frame streams to a uniform 30 fps via sliding windows.

Augmentation

Artificially multiply validation bounds using synthetic spatial rotations, coordinate jitter, and mirror reflections.

Proposed Architecture

Stage 1: Rule Heuristics

Build basic coordinate angle limits. Delivers zero training and ultra low-latency validation.

Stage 2: Classical ML

Train robust Random Forest and SVM predictors on hand-designed geometric traits.

Stage 3: Temporal DL

Introduce 1D CNN and LSTM cells to model dynamic tracking progressions over sliding cycles.

Stage 4: Graph NN

Model organic bone lengths as structural edges using high-fidelity ST-GCN graph structures.

Project Plan & Risks

TEAM MEMBER	CORE PROJECT ROLE	MILESTONE 2 ROAD MAP (NEXT 3 WEEKS)
Kunal Pandey	Data & Capture Lead	Standardize coordinate streams, assemble validation splits.
Lei Li	Machine Learning Lead	Incorporate preprocessing algorithms, baseline pipeline tests.
Chao Chen	Systems Integration Lead	Build lightweight UDP network server and input emulators.
Chenghua Jiang	Documentation & QA Lead	Audit sample dataset distribution and dynamic class balance.

CRITICAL RISKS: Keep latency under 50ms using lightweight UDP raw float arrays. Implement coordinate debouncing to eliminate false multi-triggers.

Instructor Q&A

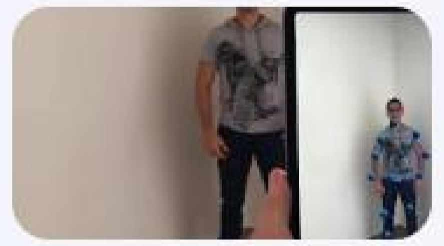
- **Benchmarking Targets:** Is our unique, self-collected gesture tracking dataset sufficient for evaluating performance?
- **Baseline Comparisons:** Is our basic heuristic threshold classifier a sufficient comparative baseline?
- **System Infrastructure:** Does the grading board favor local PC servers (Wi-Fi streams) or direct on-device Core ML execution?
- **Evaluative Metrics:** What is the formal grading weight split between accuracy, controller latency, and product design polish?
- **Game Integration:** Is a single mapped emulator game (e.g., Mario platformer) acceptable to showcase the system?

Questions & Answers

Thank you for your valuable feedback.

MotionPlay Capstone Team • Milestone 1 Check-In

Image Sources



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Source: lightbuzz.com